Alaa Richard Farghli

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EDUCATION

August 2017 – Current

Cornell University: Ithaca, NY
Doctor of Philosophy in Genetics
Advisor: Praveen Sethupathy
Committee Members: Charles Danko, Paul Soloway

September 2013 – May 2017

State University of New York at Potsdam: Potsdam, NY
Bachelor of Science in Biochemistry

RESEARCH EXPERIENCE

December 2020 – Current

Doctoral Student Researcher, Sethupathy Lab, *Biomedical Sciences, Cornell University, Ithaca, NY*
Project: Dissecting the transcriptional regulatory landscape of fibrolamellar carcinoma (FLC)

- Analyzed snATAC-seq of primary human tumor samples
- Developed custom pipeline to overlap ChRO-seq defined enhancers with snATAC-seq accessible peaks
- Elucidated FLC tumor microenvironment using snATAC-seq, snRNA-seq and spatial genomics

May 2018 – May 2020

Doctoral Student Researcher, Kwak Lab, *Molecular Biology and Genetics, Cornell University, Ithaca, NY*
Project: Detailing the transcriptional landscape of phase-separated transcription granules

- Produced preliminary results showing phase-separated granules mediate high levels of transcription
- Mentored, trained, and hosted a rotation student, and undergraduate student working on this project
- Planned and executed novel experiments to elucidate the role of phase separation in transcription
- Designed novel procedure to isolate transcription granules

May 2017 – July 2017

Undergraduate Research Student, Bou-Abdallah Lab, *Chemistry, SUNY Potsdam, Potsdam, NY*
Project: Iron accumulation inside Human Ferritin

- Designed novel characterization method to elucidate ferritin heavy/light chain ratios using capillary electrophoresis
- Characterized previously unknown iron oxidation mechanisms of ferritin using fluorimetry

September 2014 – July 2016

Undergraduate Research Student, Rossiter Lab, *Chemistry, SUNY Potsdam, Potsdam, NY*
Project: Enzyme based biosensors

- Collaborated with a research partner and associate professor in developing a biosensor to detect bacterial infections
- Successfully produced preliminary data showing promising efficacy of the biosensor
- Analyzed and quantified pig liver esterase kinetics using fluorescence techniques and MS Excel

September 2015 – May 2016

Undergraduate Research Student, *Biology, SUNY Potsdam, Potsdam, NY*
Project: Trends in Bacterial Genomic Architecture

- Implemented data mining to elucidate aspects of insertion sequence (IS) distribution, and the effect of IS element insertion in shaping modern bacterial genomes
- Identified previously unknown regions of the genome where ISs preferentially insert

June 2016 – August 2016 **Summer Research Student, Lis Lab,** *Molecular Biology and Genetics, Cornell University, Ithaca, NY*

Project: Comparative analysis of transcribed and non-transcribed enhancers

- Established CRISPR/Cas9 systems to mutate heat shock motifs present in presumed enhancers in K562 cells to better understand the roles that enhancers play in the genomic architecture

TEACHING AND MENTORING

Teaching Experience

August 2021 – September 2021 **Teaching Assistant,** *BIOMG 4400, Laboratory in Biochemistry and Molecular Biology*

- Assisted with lab setup on a weekly basis, ensuring that necessary materials and equipment were available for experiments
- Graded lab notebooks and provided constructive feedback to students, helping them to improve their scientific writing and record-keeping skills
- Held office hours to aid students and address any questions or concerns regarding the course material
- Demonstrated expertise in protein purification techniques, including salt fractionation, as well as protein analysis methods such as SDS PAGE and immunoblotting
- Assisted students in nucleic acid purifications, agarose gel electrophoresis, restriction endonuclease digestion, PCR and qPCR, DNA cloning, and DNA sequence analysis

April 2021 – April 2021 **Expand Your Horizons (EYH),** *The Magic of Coding, Cornell University*

- Instructed a cohort of six elementary school girls the fundamentals of Linux, including basic commands, file navigation and creation, to introduce them to computer programming and encourage interest in STEM

January 2020 – May 2020 **Teaching Assistant,** *BIOMG 4390, Molecular Basis in Disease*

- Held regular office hours to address student questions and concerns, aiding and helping to clarify complex concepts
- Graded exams and assignments, providing feedback to students and working closely with the course instructor to ensure consistency and accuracy in grading
- Delivered a guest lecture on a selected topic related to transcription and disease, showcasing strong understanding and expertise in the subject matter

January 2019 – May 2019 **Teaching Assistant,** *BIOMG 8310, Advanced Biochemical Methods I*

- Managed procurement of course materials for four different faculty members' sections as a teaching assistant
- Communicated regularly with each faculty member to understand their specific needs and preferences
- Ensured timely delivery of necessary materials to support faculty members' teaching

August 2019 – September 2020 **Teaching Assistant,** *BIOMG 2801, Laboratory in Genetics and Genomics*

- Conducted introductory lectures for each lab, providing students with essential information and guiding them through the experimental process
- Assisted students with conducting drosophila crosses to conduct a CRISPR-Cas9 experiment
- Graded homework and lab reports, providing constructive feedback to students and working closely with the course instructor to ensure accuracy and consistency in grading
- Held weekly office hours to address student questions and concerns, aiding and helping to clarify complex concepts

April 2018 – April 2018 **Graduate Student School Outreach Program (GRASSHOPR),** *Instructor, Geneva, NY*

- Taught biological levels of organization to a cohort of diverse elementary school students over a three-week period

- Designed and implemented lesson plans for each week, covering animal and plant cells, organs, and animals and plants respectively
- Developed strong communication and organization skills through planning and delivering effective lessons to young learners.

September 2015 – May 2017 **Teaching Assistant,** *CHEM341/342, Organic Chemistry*
- Conducted introductory lectures for each lab
- Graded lab reports and exams while providing constructive feedback to students
- Analyzed student reported results and presented findings in subsequent lab meetings

September 2015 – May 2017 **Collegiate Science and Technology Entry Program (CSTEP) Tutor**, *Organic Chemistry*
- Tutored a numerous cohorts ranging from 5-11 students, twice a week to review recent lectures
- Assisted in creating and implementing a comprehensive study guide to learn complex concepts

Course Lectures

January 2020 – May 2020 **Teaching Assistant,** *BIOMG 4390, Molecular Basis in Disease:* Transcription – Examples in Disease

Mentoring Experience

September 2021 – May 2022 Yutong Zhu, Cornell University, College of Agriculture and Life Sciences, Class of 2022 Independent Study

June 2021 – August 2021 Simone Alma Evans, University of Maryland**,** Cell Biology and Molecular Genetics, Summer REU student

June 2019 – August 2019 Erica M. Rosario, City University of New York, Hunter College, Biology, Summer REU student

October 2018 – December 2018 Hector Loyola Irizarry, Cornell University, Biological and Biomedical Sciences, Rotation Student

June 2018 – August 2018 Adriana Velez, University of Puerto Rico**,** Cellular and Molecular Biology, Class of 2019, Summer REU student

PROFESSIONAL DEVELOPMENT

December 2022 **Computational Genomics Course,** *Cold Spring Harbor Laboratory*
August 2022 **Summer Success Symposium,** *Cornell University, Ithaca, NY*
August 2019 **Summer Success Symposium,** *Cornell University, Ithaca, NY*

PUBLICATIONS

1) Liu. S., Benito-Martin. A., Vatter, F., Hanif, S., Liu, C., Bhardwaj, P., Sethupathy, P., **Farghli, A.,** Piloco, P., Paik, P., Mushannen, M., Otterburn, D., Cohen, L., Bareja, R., Krumsiek, J., Cohen-Gould, L., Calto, S., Spector, J., Elemento, O., Lyden, D., Brown, K., (2023). Breast adipose tissue-derived extracellular vesicles from women with obesity stimulate mitochondrial-induced dysregulated tumor cell metabolism. *Nature Cancer, (*In review)

2) Francisco, A. B., Li, J., **Farghli, A. R.**, Kanke, M., Shui, B., Munn, P. R., Grenier, J. K., Soloway, P. D., Wang, Z., Reid, L. M., Liu, J., & Sethupathy, P. (2022). Chemical, Molecular, and Single-nucleus Analysis Reveal Chondroitin Sulfate Proteoglycan Aberrancy in Fibrolamellar Carcinoma. *Cancer Research Communications*, *2*(7), 663–678. https://doi.org/10.1158/2767-9764.CRC-21-0177

RESEARCH PRESENTATIONS

May 2023 **Alaa R. Farghli**, Bo Shui, Seoyeon Lee, Paul Munn, Jennifer Grenier, Paul Soloway, David McKeller, Allison Kirk, Paul Thomas, Praveen Sethupathy. *Integrative single-nucleus and spatial analysis maps the tumor environment of fibrolamellar carcinoma*, Gordon Research Symposium, oral presentation

May 2023	**Alaa R. Farghli**, Bo Shui, Seoyeon Lee, Paul Munn, Jennifer Grenier, Paul Soloway, David McKeller, Allison Kirk, Paul Thomas, Praveen Sethupathy. *Integrative single-nucleus and spatial analysis maps the tumor environment of fibrolamellar carcinoma*, Gordon Research Conference, poster presentation
October 2022	**Alaa R. Farghli**, Bo Shui, Seoyeon Lee, Paul Munn, Jennifer Grenier, Paul Soloway, Praveen Sethupathy. *Decoding the tumor biology of the orphan malignancy fibrolamellar carcinoma at single-cell resolution*, BMCB-GGD Symposium, poster presentation
April 2022	**Alaa R. Farghli**, Bo Shui, Seoyeon Lee, Paul Munn, Jennifer Grenier, Paul Soloway, Praveen Sethupathy. *Decoding the tumor biology of the orphan malignancy fibrolamellar carcinoma at single-cell resolution*, American Association for Cancer Research, poster presentation
August 2019	**Alaa Farghli**, Hojoong Kwak. *Discerning the transcriptional landscape of phase separated transcription bodies*, Cold Spring Harbor Laboratory, Mechanisms of Eukaryotic Transcription, poster presentation

OUTREACH, DIVERSITY, AND INCLUSION

Leadership

March 2021 – Current

President, Arab Graduate Student Association (AGSA), Cornell University, Ithaca, NY
- Founded AGSA to address the social, academic, and cultural needs and concerns of graduate and professional students of the Arab community at Cornell University.
- Organized discussions on colorism across communities of color with the Black Graduate and Professional Student Association (BGPSA) and Society for the Advancement of Chicanos/Hispanics and Native Americans in science (SACNAS).
- Developed and implemented monthly "Arab in Academia" series, connecting graduate and professional students with Arab faculty members at Cornell and promoting representation and visibility in academia
- Coordinating with North African, Mediterranean, and Middle Eastern organizations to expand the national bone marrow registry by encouraging their members to donate samples, as part of a collaborative effort with the nonprofit organization Marrow Mates

January 2022 – Current

Center for Vertebrate Genomics (CVG) Trainee Executive Committee
- Organized semesterly "CVG Stories" where CVG faculty members provide informally share with attendees about their life journey, development of academic interests, career trajectory, leadership experiences, etc.
- Updated the CVG twitter page weekly

September 2021 – Current

Office of Inclusion and Student Engagement (OISE) Graduate Coordinator for Community Engagement
- Organized a range of social and academic events to enhance the Cornell community's sense of inclusion and belonging

June 2021 – August 2021

Research Experience for Undergraduates (REU), Molecular Biology and Genetics
- Acted as Peer Mentor Organizer for Cornell MBG's Research Experience for Undergraduates (REU).
- Increased interactions between undergraduates and graduate students by organized weekly social events between undergraduate peer mentees and graduate student peer mentors.

December 2021 – March 2022

Genetics, Genomics and Development Admissions Committee, Cornell University, *Student Representative*
- Reviewed dozens of graduate student applicants and provided a student perspective in the holistic admissions process

June 2019 – August 2019 **Molecular Biology and Genetics National Science Foundation (NSF) Research Experience for Undergraduates (REU) organizer,** *Cornell University, Ithaca, NY*
- Organized weekly social and professional events for a cohort of 10 summer research students

September 2018 – May 2019 **Health Student Association Food Drive Organizer**
- Participated in weekly food drives with Health Student Association (HSA) to address food insecurity
- Communicated and collected food from various local grocery stores to distribute to the general student body.

October 2018, 2020 **BMCB-GGD Symposium Organizer**
- Served as the Treasurer for the student-run symposium, responsible for managing and allocating funds for the event, as well as overseeing the collection of payments from participants and sponsors
- Played an instrumental role in the successful transition of a symposium from an in-person event to a virtual format during the pandemic

March 2018 – December 2019 **Molecular Biology and Genetics Diversity Council**
- Participated in the MBG (Molecular Biology and Genetics) Diversity Council, with a focus on increasing retention of graduate students, and applications from people from diverse backgrounds.
- Founded and coordinated the Genetics, Genomics and Development “Buddy System” by recruiting and pairing year two and beyond graduate students to act as mentors to first year graduate students.

Outreach

August 2022 **Consider Cornell Graduate Diversity Recruitment Program,** Cornell University
- Served as a panelist to speak and provide a first-hand overview of graduate education and admissions process with prospective Cornell graduate students

October 2022 **SACNAS National Diversity in STEM,** Puerto Rico
- Represented the graduate school and spoke with prospective graduate students and promoting the program to a diverse audience

June 2021 – January 2022 **Vaccination Conversations with Scientists (VaCS)**
- Participated in multiple in-person outreach events in Tompkins County to address vaccine hesitancy in local community.

October 2021 **SACNAS National Diversity in STEM**
- Represented the graduate school and spoke with prospective graduate students and promoting the program to a diverse audience

September 2019 – August 2019 **Research Experience for Undergraduates (REU), Molecular Biology and Genetics**
- Participated as a Peer Mentor for Cornell MBG’s Research Experience for Undergraduates (REU).

December 2017 – Current **Graduate Student Ambassador, Cornell University, Ithaca, NY**
- Serve as a panelist for various events to speak with prospective Cornell graduate students

October 2016 **State University of New York, University at Potsdam**

Organized the fifth annual “Chemtoberfest!” as Chemistry Club President, our largest ever, attracting over 1000 students from surrounding schools and students from SUNY Potsdam